

Answer all questions.

1. A new timer, in the shape of a ball, has been produced. When it is released it starts a built-in stop clock. When the ball next hits something the timer stops. The time taken is displayed on its screen.



A teacher uses the ball timer for an experiment. He holds the ball out of his third floor classroom window and then carefully releases it. The time taken to fall is recorded in a table. The ball is dropped 5 times during the experiment.

Drop number	1	2	3	4	5
Time (s)	1.48	1.10	1.52	1.54	1.46

- (a) Explain how the teacher could find out if the results are reproducible. [2]
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- (b) **Circle** the result in the table which is most likely to be anomalous. [1]
- (c) Calculate the mean time for the ball timer to fall. [2]

Mean time to fall = s



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- (d) The mean impact velocity of the ball on the ground is 14.2 m/s.
Use the equation:

$$x = \frac{u + v}{2} t$$

to calculate the drop height of the ball.

[2]

Drop height = m

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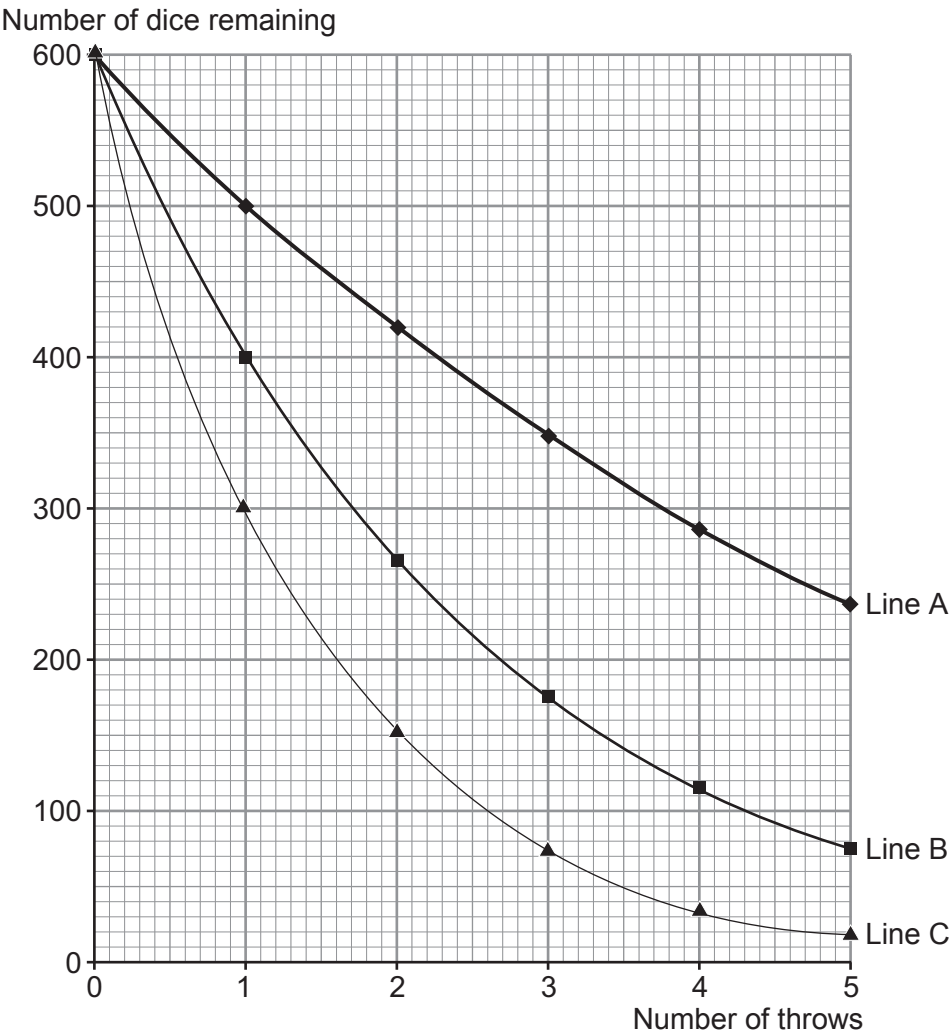


5. A physics teacher uses 600 identical red dice with her class to model the decay of different radioisotopes. To collect her first set of data she throws the 600 dice. She then removes all of the **sixes** that are face up. She then counts the remaining dice and repeats the process for a total of 5 throws.

In her second demonstration she repeats the experiment but removes all of the **fives** and **sixes** that are face up.

In her third demonstration, she removes all of the **fours**, **fives** and **sixes** that are face up.

She records her data on a spreadsheet. This quickly produces a graph for her class to see. The graph is shown below.



(a) (i) Explain why the teacher chooses to use a large number of dice for her demonstration. [2]

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- (ii) Using only information from the graph tick (✓) the **three** correct statements. [3]
Space for working:

Line A represents the data from the third demonstration. ☐

Line B shows the equivalent of three half-lives in 5 throws. ☐

Line C could also represent the same demonstration modelled with 600 coins. ☐

Line A represents the radioisotope that decays at the quickest rate. ☐

Line B has an activity that is $\frac{1}{8}$ of its original after 5 throws. ☐

Line C represents the longest half-life. ☐

- (iii) The teacher suggests to her class that an improved model of radioactive decay would be to include 20 blue cubes, along with the 600 red dice, at the start. The blue cubes **would not** be removed during the demonstration but would be counted each time.

I. State what physical quantity the blue cubes represent. [1]

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II. State how the presence of the blue cubes affects **line A** on the graph. [1]

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(b) Strontium-93 (Sr-93) is a radioactive source which decays with a half-life of 8 minutes.

- (i) A sample of Sr-93 has a mass of 68.0 mg. Calculate the mass of undecayed Sr-93 left after 56 minutes. Show your working and give your answer to **1 significant figure**. [3]

Mass = mg

- (ii) A scientist works with Sr-93 which decays by beta and gamma emission. When he is not using the Sr-93, explain why, for his own safety, he has to store it in a lead box. [3]

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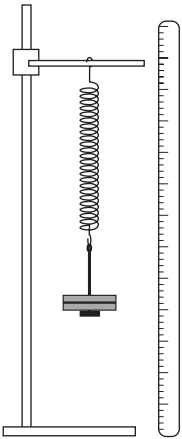
7. A Welsh company has been asked to design a toy that is able to launch a model plastic plane. They decide to use a spring mechanism in its design. Initially they carry out an experiment to investigate whether a spring is suitable for use in the toy.

Slotted masses are added to exert a force on the spring.

The resulting length of the spring is measured.

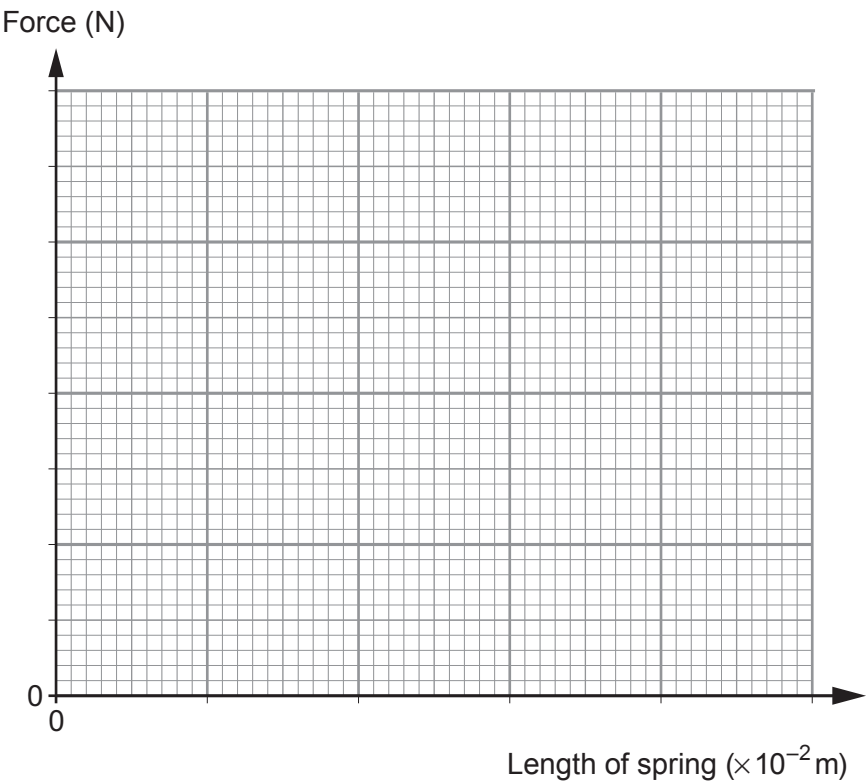
The results are shown in the table below.

Mass (g)	Force (N)	Length of spring ($\times 10^{-2}$ m)
100	1.0	7.5
200	2.0	10.0
300	3.0	12.5
600	6.0	20.0
700	7.0	22.5



- (a) (i) Plot the data on the grid below and draw a suitable line.

[4]



- (ii) State the length of the spring when no force is applied.

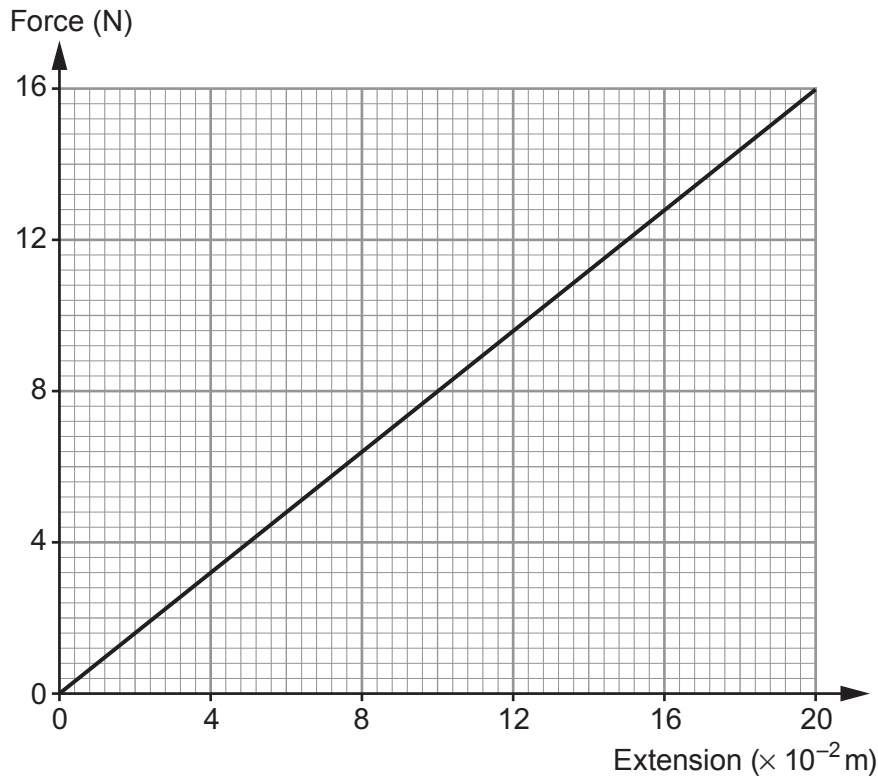
[1]

Length of spring = $\times 10^{-2}$ m



- (b) The design company decides to connect two of these springs in parallel with each other in the launch mechanism of the model plastic plane.

The graph shows how the extension of this spring system changes with force.



The two spring system is stretched and a model plastic plane of mass 55×10^{-3} kg is attached. A trigger is pressed, which releases one end of the spring system, and the plane is launched.

- (i) The two spring system is stretched to a maximum extension of 15×10^{-2} m. Use the graph and an equation from page 2 to calculate the work done stretching this spring system. Give an appropriate unit with your answer. [4]

Work done =

Unit



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- (ii) Health and safety guidelines limit the launch velocity of the plastic plane to a maximum of 4 m/s. Using an equation from page 2 and your answer from (b)(i), calculate the launch velocity of this plane (mass = 55×10^{-3} kg) **and state** whether the launch system design meets this safety guideline. [3]

Launch velocity = m/s

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- (iii) Explain why the actual launch velocity is always less than your calculated launch velocity. [2]

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- (iv) A student suggests that a maximum momentum value would be more appropriate than stating a maximum velocity restriction for the launch. Explain why this would be an improvement to the health and safety guidelines. [2]

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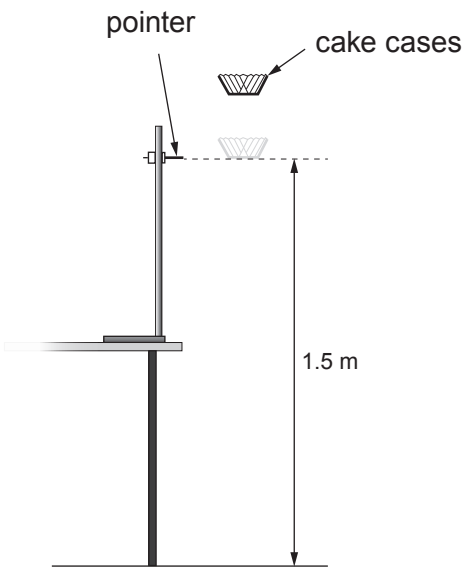
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4. A group of students investigate how the terminal speed of falling paper cake cases depends on their mass. They use the apparatus shown in the diagram below.



The cake cases are dropped from at least 20cm above the pointer. This allows them to reach a terminal speed by the time they reach the pointer. The time taken for the cases to fall the distance of 1.5 m between the pointer and the floor is measured.

The students are given a blank table to record their results. This is shown below.

Number of cake cases	Mass of cake cases (g)	Time taken for paper case cases to fall (s)						Terminal speed (m/s)
		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	Mean	
1								
2								
3								
4								
5								

- (a) (i) The same cake cases are used throughout the experiment. State the other controlled **variables** in the experiment. [1]

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(ii) Explain the advantage of 5 trial timings for each number of cases. [2]

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(iii) The students are told the mean mass of a paper case. Explain how the quality of results collected is affected if the cases used each time are weighed rather than calculated by using the mean mass of a case. [2]

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(b) (i) Explain, in terms of forces, how the cake cases reach a terminal speed. [2]

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(ii) Describe how the experiment could be extended to check that the cases are falling with a terminal speed over the 1.5m shown. [2]

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- (c) The cake cases used are supplied in a box containing 250 cases. The combined mass of the box **and** cases is 118 g. The mass of the empty box is 18 g. The terminal speed of 4 paper cases is found to be 2.54 m/s.
The weight of a 1 kg mass on Earth is 10 N.

Use equations from page 2 to calculate:

- (i) the time taken for the 4 paper cases to fall the 1.5 m shown in the diagram on page 9. [2]

Time = s

- (ii) the air resistance acting on 4 paper cases when they fall at terminal speed.
Show your working. [4]

Air resistance = N

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2. A group of students investigate how the surface area of a falling paper cake case affects its terminal speed.

- Cake case 1 has a mass of 0.5 g and a surface area of 100 cm².
- Cake case 1 is dropped from a height of 1.80 m but only timed over the final 1.50 m of the fall.

The students’ results are shown in the table below.

Drop time (s)			Mean drop time (s)	Drop distance (m)
Attempt 1	Attempt 2	Attempt 3		
0.96	0.92	0.94		1.50

(a) (i) The students decide there are no anomalies. Explain why. [1]

(ii) **Complete the table** to show the mean drop time. [1]
Space for calculation.

(b) The experiment is repeated with cake case 2.
It has the same shape and the same mass as cake case 1.
However, cake case 2 has a surface area of 50 cm².
The students correctly calculate the terminal speed for both cake cases.

Cake case 1		
Mass (g)	Surface area (cm ²)	Terminal speed (m/s)
0.5	100	1.6

Cake case 2		
Mass (g)	Surface area (cm ²)	Terminal speed (m/s)
0.5	50	2.3



- (i) A cake case reaches terminal speed when its weight is balanced by air resistance. **Tick (✓) the three correct statements.** [3]

- Cake case 2 has the same terminal speed as cake case 1. ☐
- Cake cases 1 and 2 have identical weight. ☐
- At terminal speed, cake case 1 experiences a greater value of air resistance than cake case 2. ☐
- At terminal speed, both cake cases experience identical values of air resistance. ☐
- At terminal speed, cake case 1 experiences a smaller value of air resistance than cake case 2. ☐
- At terminal speed, both cake cases have zero acceleration. ☐

- (ii) Before the experiment was carried out the students made the following prediction:
“If the surface area of the cake case is halved its terminal speed will double.”
Use data from the tables on the previous page to explain whether their prediction was correct. [2]

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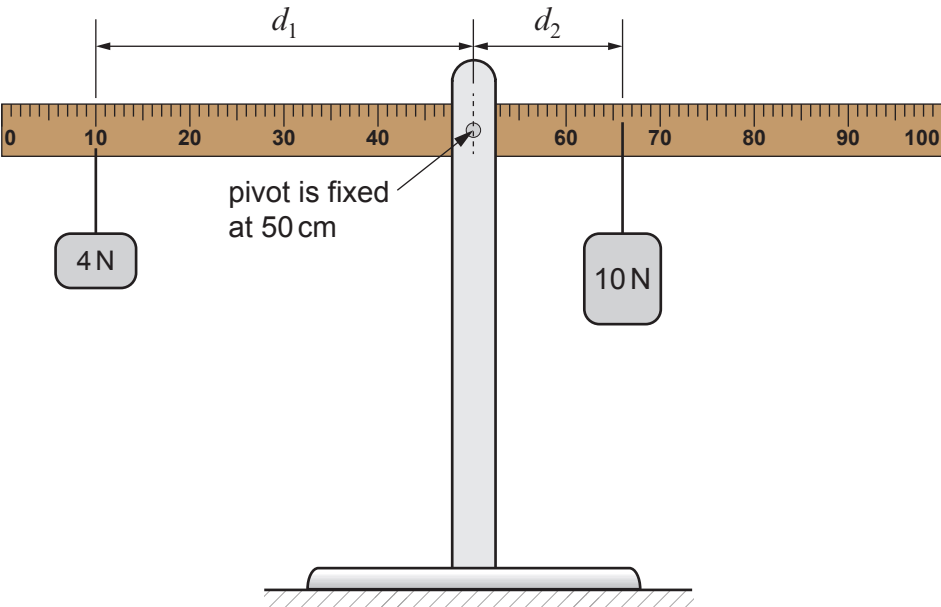
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3. Mary investigated moments using a **100 cm** ruler and two different weights.

The apparatus used is shown below.



Mary checked the ruler was horizontal. She then attached weights to the ruler. The 4 N weight, W_1 , was attached to the ruler a distance, d_1 , from the pivot. The 10 N weight, W_2 , was then attached to the ruler to make it horizontal again. The distance, d_2 , of the 10 N weight from the pivot was noted.

Mary's results are shown in the table.

Weight, W_1 (N)	Distance, d_1 (cm)	Weight, W_2 (N)	Distance, d_2 (cm)
4	40	10	16
4	35	10	14
4	20	10	8
4	15	10	6
4	5	10	2

(a) Mary states the resolution of the ruler she used in this experiment as 1 cm. Use information from the diagram to explain whether Mary is correct.

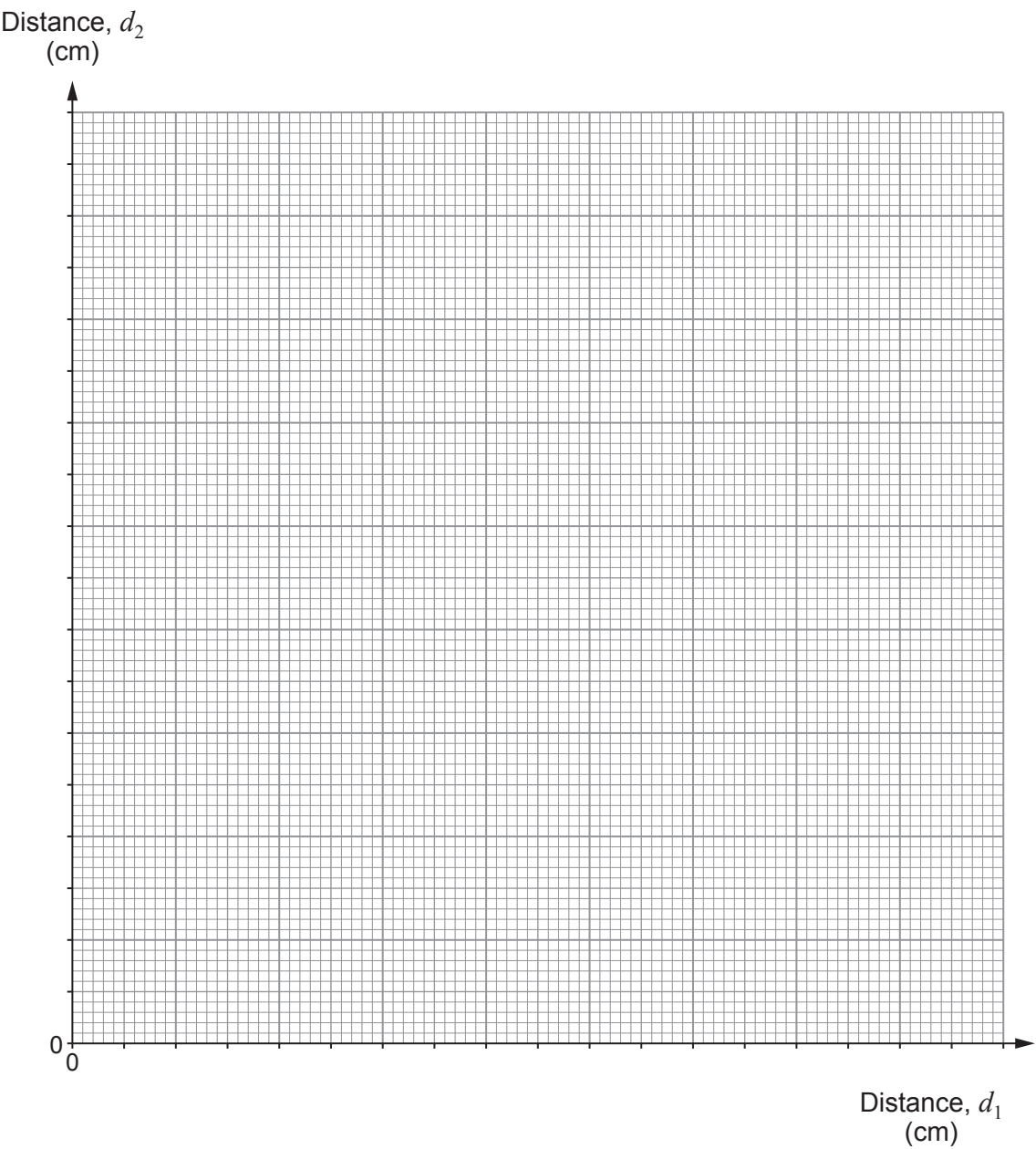
[1]



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(b) (i) Plot the data on the grid below and draw a suitable **straight** line.

[3]



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(ii) Mary suggests that the value of the gradient of the graph is the same as $\frac{W_1}{W_2}$.

Use data from the graph and the table to explain whether Mary is correct. [3]

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(c) (i) Mary now places the 10 N weight at a distance, d_2 , of 32 cm from the pivot.
Use an equation from page 2 to calculate its clockwise moment about the pivot. [2]

Clockwise moment =Ncm

(ii) Explain, using moments, why the ruler cannot now be balanced using the 4 N weight. [2]

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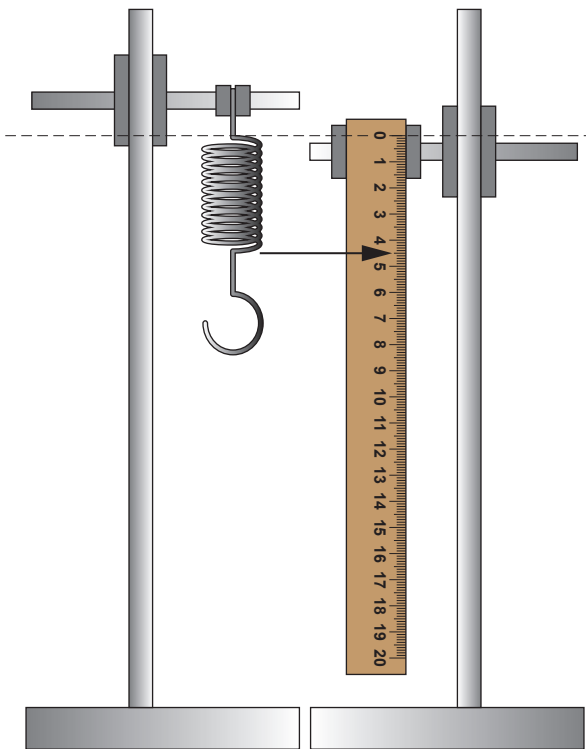
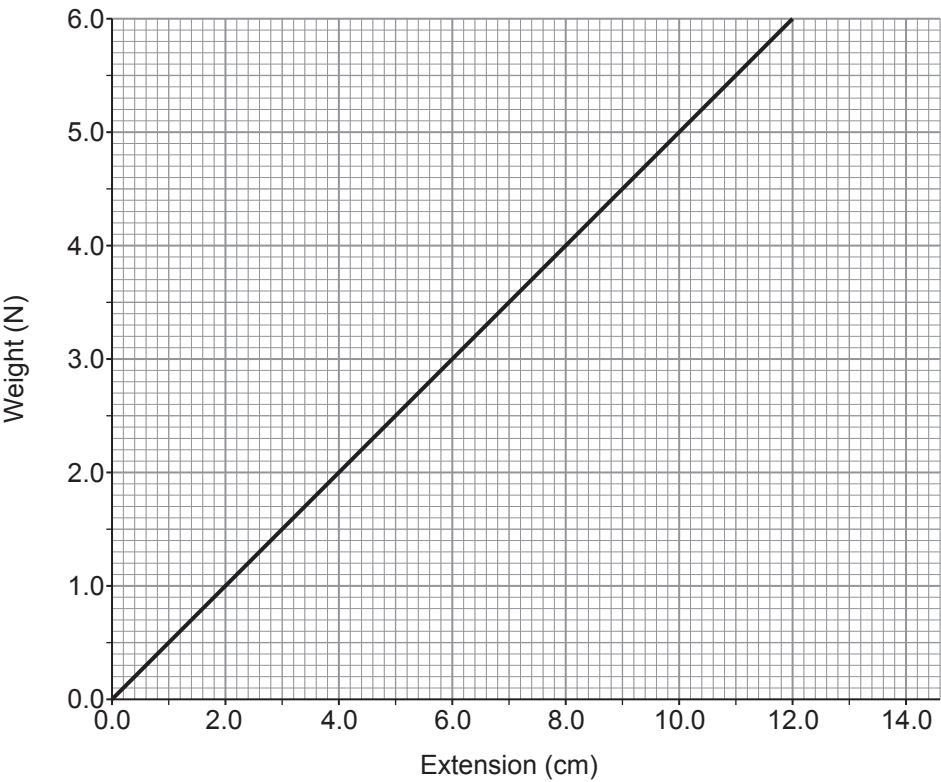
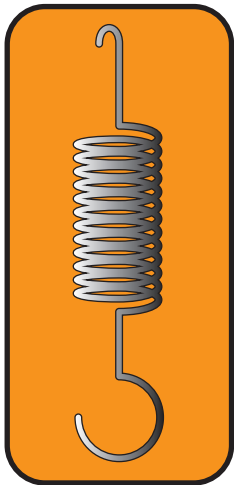
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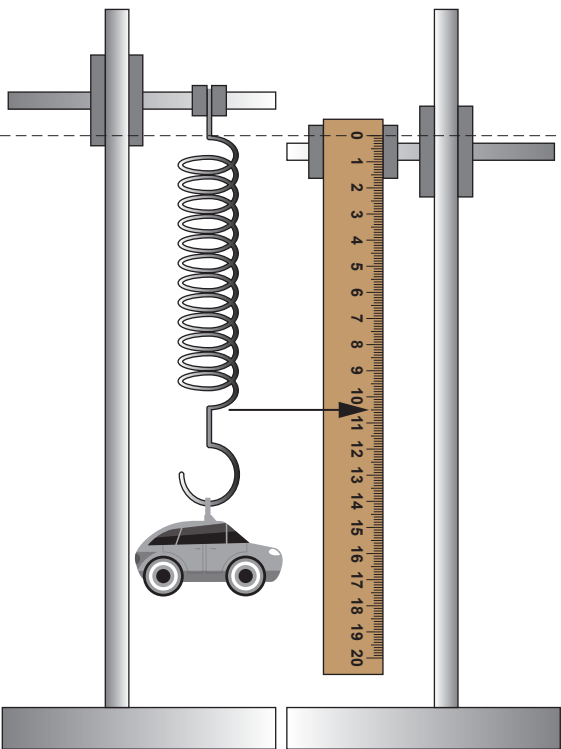


8. A school science technician bought a spring for an experiment.
The data leaflet about the spring is shown below.

Data leaflet



Picture 1



Picture 2



- (a) A student used the spring to determine the mass of a toy.

Two pictures of the experiment were taken by the student.
Picture 1 shows the unstretched spring and Picture 2 shows the spring with the toy attached.

- (i) Use information from the two pictures to determine the spring's extension with the toy attached. [1]

spring's extension = cm

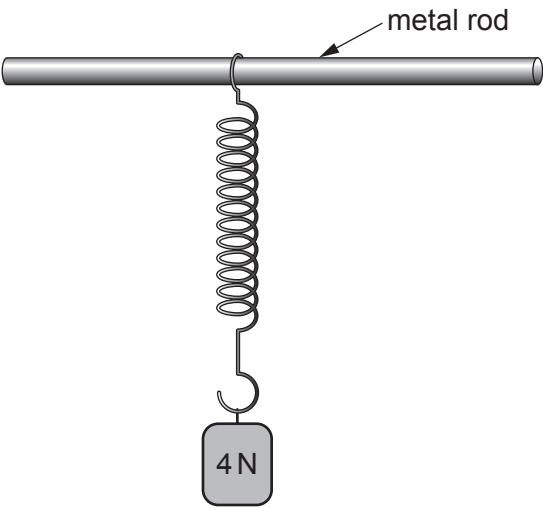
- (ii) Use information from the leaflet, and an equation from page 2, to calculate the mass of the toy. ($g = 10 \text{ N/kg}$) [3]

mass of toy = kg

- (b) (i) State Newton's 3rd law. [2]

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- (ii) The diagram below shows the spring suspended from a metal rod.
A weight of 4 N is attached to the spring.



State the **size and direction** of the force that the spring applies to the weight. [2]

size = N direction =



4. A class investigated radioactive decay using 50 dice, each with 8 sides.



They rolled the dice and removed any which landed with an 8 facing upwards, to represent a decayed nucleus.

They recorded the number of dice remaining.

They rolled the remaining dice and again removed any which landed with an 8 facing upwards.

This was repeated until they had rolled the dice 10 times in total.

Some of the results are shown in the table below.

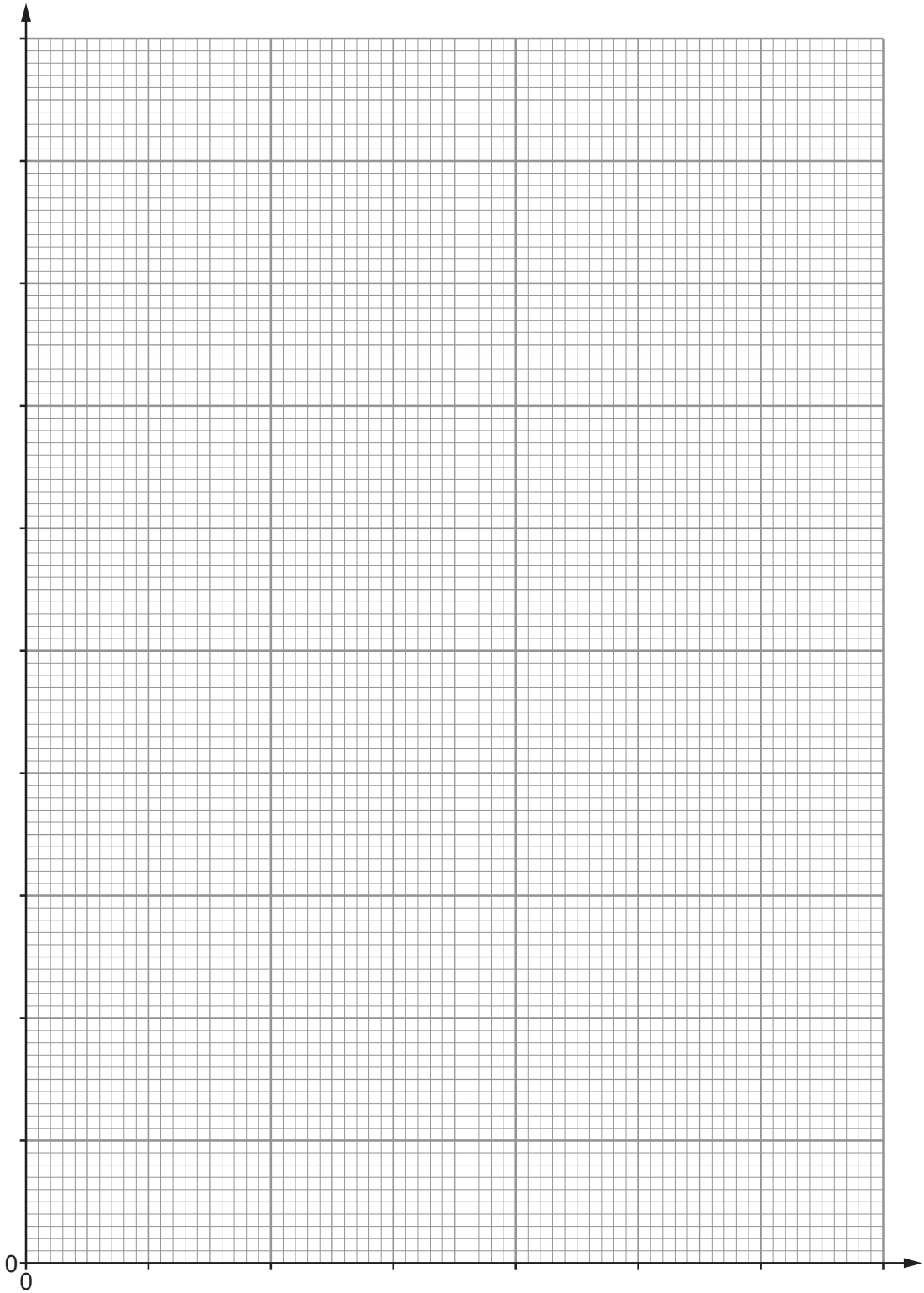
Number of throws	Number of dice remaining
0	50
2	37
4	29
6	23
8	18
10	15

(a) (i) Plot the data on the grid opposite and draw a suitable line. [3]



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Number of dice remaining



Number of throws



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(ii) Use your graph to determine how many dice were removed on the **3rd throw**. [2]

Number of dice removed =

(iii) Determine how many throws it took for the initial number of dice to halve.
Show how you obtained your answer on the graph. [2]

Number of throws =

(iv) If the experiment was repeated with 1 000 dice, use your results to predict how
many throws it would take to reduce them to 125. [2]

Number of throws =

(b) Explain why it is preferable to use a larger sample size in this experiment. [2]

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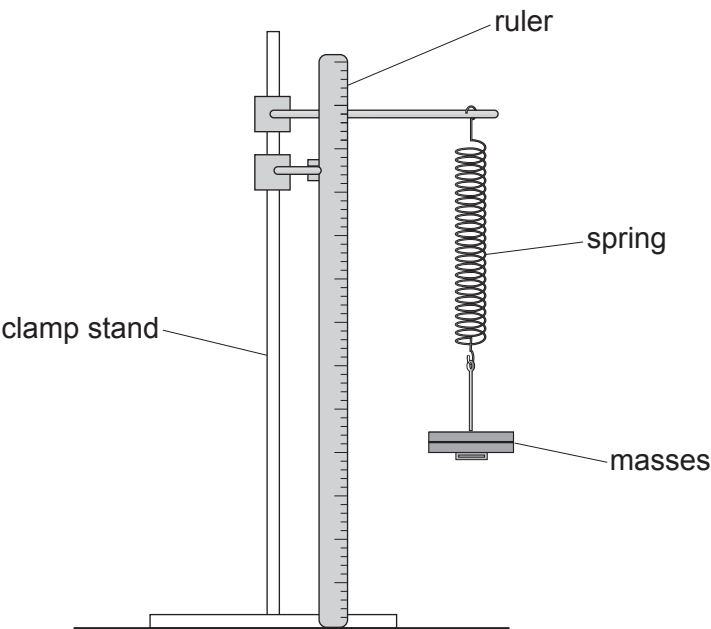
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5. Some students carry out an experiment to determine the relationship between the force applied to a spring and its extension. They use the following apparatus.



They collect data by loading and then unloading the spring. Their results are shown in the table below.

Mass (g)	Force (N)	Extension (cm)	
		Loading	Unloading
0	0	0.0	0.0
100	1	4.4	4.2
200	2	8.9	9.3
300	3	13.6	13.7
400	4	18.2	18.5
500	5	22.5	22.5

- (a) The uncertainty in extension is given by:

uncertainty in extension =
$$\frac{\text{maximum value of extension} - \text{minimum value of extension}}{2}$$

- (i) State the mass with the greatest uncertainty in extension. [1]

value of mass = g



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(ii) Calculate this uncertainty. [2]

uncertainty = cm

(b) Suggest how the students could adapt the apparatus to reduce the uncertainty in their results. [1]

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(c) Explain how they could use **all** of the results to determine a mean value for the spring constant. [3]

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(d) The students determine the spring constant to be 22.3 N/m.
The true value of the spring constant is 22.2 N/m.
Explain what this tells the students about their data. [2]

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